

**TED UNIVERSITY**

Department of Computer Engineering
CMPE453: Embedded Systems, Midterm EXAM

Sample

Duration: 120 minutes

FULL NAME :

STUDENT ID :

SIGNATURE :

QUESTIONS	POINTS	
Q1		/25
TOTAL		/100

Please Read Carefully

- This is a closed-book exam.
- Talking is strictly forbidden during the exam.
- Use of electronic devices including mobile phones is forbidden during the exam.
- Students will not be allowed to enter 10 minutes after the exam starts and will not be able to leave in the last 10 minutes of the exam.
- Exchange of items (erasers, pens, pencils, etc.) during the exam is strictly forbidden.
- There should be total of 4 questions. Check your exam paper at the beginning.
- Show your work clearly in the spaces provided.

Q1. Programming AVR for digital input and output:

- a)
- [5 pts]**
- What is the value of Port B after the execution of the following statements?

```
PORTB = 0b01010101;
```

```
PORTB = PORTB | 0b00001000;
```

Solution: `PORTB = 0b01010101`

- b)
- [5 pts]**
- Write the AVR C code which will reset (clear) the bit 3 (4
- th
- bit) without disturbing the rest of the pins of Port B.

Solution:

The mask for clearing bit 3 is `0b11110111` (all bits are `'1'` except bit 3, which is `'0'`).

Therefore, code to Clear Bit 3 is:

```
PORTB &= ~(1 << 3);
```

or

```
PORTB &= 0b11110111;
```

or

```
PORTB = PORTB & 0b11110111;
```

- c)
- [15 pts]**
- Suppose an AVR microcontroller is properly connected with necessary registers to a push button with pin 3 of port B and to a seven segment display with pin 1, 2, 3, and 4 for port D. Also consider that the function
- void displayDigit(unsigned char i)*
- turns on necessary LEDs of the seven segment display to display digit
- i*
- in Port D. Now program the microcontroller to setup
- only the necessary pins**
- of the relevant I/O ports to count the number of pushes in the button and display the corresponding decimal digits (0 - 9) using the seven segment display. At the beginning, it displays 0, and when the button is first pressed it will display 1 and keep displaying 1 until the button is pressed again. Upon the next press, it will display 2 and continue until the number reaches 9. In the next press of the button, it will displaying 0 again. (Note: The pin index of a port start with 0.)

Solution:

```
#include <avr/io.h>
#include <util/delay.h>
int main(void) {
    // ----- Inits ----- //
    PORTB |= (1 << PB3);
    DDRC = 0b00111110;
    bool old_state = false;
    bool current_state;
    // ----- Event ----- //
    unsigned char count = 0;
    displayDigit(count);
    while(1)
    {
        current_state = !(PINB & (1<<PB3))    //bit_is_clear(PINB, PB3);
        if(current_state && !old_state) {
            displayDigit(count++);
            count = count%10;
        }
        old_state = current_state;
    }
    return 0;
}
```